PAPER_714: UQFF Knowledge Base 18: THz Q-Scope Signal Analysis (Set 50: Signals 41-50)

Class: `UQFFKnowledgeBaseKB18` **CP4 Entry:** #298 **Keywords:** THz, q-scope, signals 41-50, oscilloscope, 200ns, 500mV, flow state, UQFF KB18 **Session:** 175 | **Version:** v5.32 **Source:** grok_share_ba508f76c8e.txt

Abstract

UQFF Knowledge Base 18 (KB18) presents the detailed analysis of THz q-scope Set 50 (Signals 41-50), captured at 16:48:23–16:50:20 UTC-4 on the Star-Magic q-scope apparatus. All 10 signals exhibit 1.246 THz fundamental with the characteristic Ch1/Ch2 dual-channel signature used to classify ACE/DCE flow state.

1. Instrument Configuration (Set 50)

Parameter	Value
Time/Div	200 ns
Volt/Div	500 mV
Channels	Ch1 (sensor A), Ch2 (sensor B)
Duration	117 s (16:48:23–16:50:20)
Signals	41–50 (10 total)

2. Signal Amplitude Data

Signal	Ch1 (V)	Ch2 (V)	Flow State
41	0.60	0.35	Normal
42	0.65	0.40	Normal
43	0.60	0.35	Normal
44	0.55	0.30	Chaotic
45	0.50	0.35	Inverted
46	0.60	0.40	Inverted
47	0.55	0.35	Chaotic
48	0.50	0.30	Chaotic
49	0.50	0.35	Inverted
50	0.50	0.35	Inverted

3. UQFF Signal Energy

$$E_{signal}(i) = \frac{V_{eff,i}^2}{Z}\,\Delta t, \quad Z = 50\,\Omega, \quad V_{eff} = V_{ch1}/\sqrt{2}$$

Bundle sum (weighted by flow state):

$$B_{set50} = \sum_{i:fs\neq 0} E_i \cdot fs(i)$$

4. U_m Contribution

$$U_{m,i} = \mu_J\,\omega_{THz}\,V_{ch1,i}\left(1 - e^{-\kappa(t+t_i)}\right)$$

References

- Q-Scope Laboratory Records Set 50, Star-Magic 2026
- UQFF Framework v5.32, Star-Magic Session 175

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